

Characterizing States of Consciousness Using Compression-Spectrum-Derived Features from ECoG Recordings

Feature Extraction and Statistical Analysis

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1 Overview

The following features were extracted from the experiment recordings:

- ETC value
- Spectrum Bandwidth
- Max Scale
- Mean Fluctuation
- Mean Entropy

2 Experimental Structure

Each experimental condition shares the following hierarchical recording structure:

Unit	Description
1 recording	10 bins
1 bin	2 seconds
1 second	200 data points

3 Ketamine (KTMD) Experiment — Monkey 1 (“Chibi”)

3.1 KTMD Day 1

Total: 2 sessions $\times$ 2 recordings/session = 4 recordings.

Table 1: KTMD Day 1 — Feature Means $\pm$ Standard Deviations

Recording	ETC (mean $\pm$ SD)	Bandwidth (mean $\pm$ SD)	Max Scale (mean $\pm$ SD)	Fluctuation (mean $\pm$ SD)	Mean Entropy (mean $\pm$ SD)
Recording 1	25.17 $\pm$ 3.29	10.81 $\pm$ 1.63	27.70 $\pm$ 14.01	5.27 $\pm$ 2.76	0.05 $\pm$ 0.08
Recording 2	25.38 $\pm$ 3.19	10.80 $\pm$ 1.65	27.32 $\pm$ 12.52	5.00 $\pm$ 2.22	0.05 $\pm$ 0.08
Recording 3	27.62 $\pm$ 3.69	10.47 $\pm$ 1.61	20.77 $\pm$ 10.90	4.00 $\pm$ 2.12	0.12 $\pm$ 0.12
Recording 4	28.26 $\pm$ 3.09	10.49 $\pm$ 1.49	18.09 $\pm$ 7.80	3.51 $\pm$ 1.48	0.14 $\pm$ 0.12

3.2 KTMD Day 2

Total: 2 sessions $\times$ 2 recordings/session = 4 recordings.

4 Sleep (SLP) Experiment — Monkey 1 (“Chibi”)

4.1 SLP Day 1

Total: 3 sessions $\times$ 2 recordings/session = 6 recordings.

Table 2: KTMD Day 2 — Feature Means $\pm$ Standard Deviations

Recording	ETC (mean $\pm$ SD)	Bandwidth (mean $\pm$ SD)	Max Scale (mean $\pm$ SD)	Fluctuation (mean $\pm$ SD)	Mean Entropy (mean $\pm$ SD)
Recording 1	24.92 ± 3.27	10.52 ± 1.67	28.71 ± 12.18	5.29 ± 2.28	0.05 ± 0.08
Recording 2	26.18 ± 3.44	10.55 ± 1.57	24.42 ± 11.63	4.66 ± 2.22	0.08 ± 0.11
Recording 3	27.95 ± 3.63	10.32 ± 1.60	20.32 ± 12.20	3.86 ± 2.40	0.13 ± 0.13
Recording 4	28.26 ± 3.46	10.40 ± 1.59	18.74 ± 8.30	3.56 ± 1.62	0.14 ± 0.12

Table 3: SLP Day 1 — Feature Means $\pm$ Standard Deviations

Recording	ETC (mean $\pm$ SD)	Bandwidth (mean $\pm$ SD)	Max Scale (mean $\pm$ SD)	Fluctuation (mean $\pm$ SD)	Mean Entropy (mean $\pm$ SD)
Recording 1	25.61 ± 2.56	11.50 ± 1.48	20.58 ± 6.98	4.05 ± 1.42	0.07 ± 0.07
Recording 2	26.23 ± 2.65	11.34 ± 1.44	19.80 ± 6.74	3.86 ± 1.43	0.09 ± 0.09
Recording 3	25.67 ± 2.72	11.52 ± 1.44	21.32 ± 8.22	4.26 ± 1.72	0.08 ± 0.08
Recording 4	26.35 ± 3.06	11.17 ± 1.50	21.45 ± 9.99	4.26 ± 2.04	0.09 ± 0.09
Recording 5	26.53 ± 2.93	11.19 ± 1.46	21.16 ± 10.19	4.29 ± 2.10	0.09 ± 0.09
Recording 6	26.38 ± 2.80	11.23 ± 1.43	20.59 ± 7.71	4.15 ± 1.74	0.09 ± 0.09

4.2 SLP Day 2

Total: 3 sessions $\times$ 2 recordings/session = 6 recordings.

Table 4: SLP Day 2 — Feature Means $\pm$ Standard Deviations

Recording	ETC (mean $\pm$ SD)	Bandwidth (mean $\pm$ SD)	Max Scale (mean $\pm$ SD)	Fluctuation (mean $\pm$ SD)	Mean Entropy (mean $\pm$ SD)
Recording 1	25.79 ± 2.77	11.35 ± 1.49	21.91 ± 8.56	4.28 ± 1.69	0.07 ± 0.08
Recording 2	25.35 ± 2.64	11.49 ± 1.48	22.41 ± 7.97	4.41 ± 1.59	0.06 ± 0.07
Recording 3	26.10 ± 2.52	11.42 ± 1.46	19.74 ± 6.70	3.82 ± 1.26	0.08 ± 0.08
Recording 4	25.74 ± 2.67	11.44 ± 1.49	21.81 ± 7.99	4.19 ± 1.55	0.07 ± 0.08
Recording 5	26.75 ± 2.81	11.09 ± 1.45	20.82 ± 9.48	4.12 ± 1.83	0.09 ± 0.08
Recording 6	26.53 ± 2.85	11.16 ± 1.45	20.16 ± 7.66	4.05 ± 1.63	0.10 ± 0.09

5 Averaged State Comparison (15 Recordings per State)

Each state average is computed over 15 recordings, where:

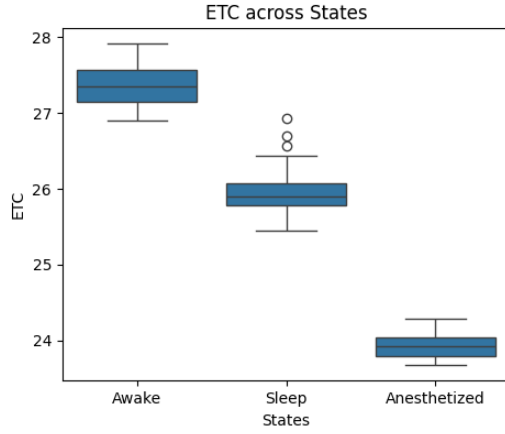
1 recording $\rightarrow$ 10 bins $\rightarrow$ 1 bin = 2 sec $\rightarrow$ 1 sec = 200 data points

Table 5: State-level Feature Means $\pm$ Standard Deviations

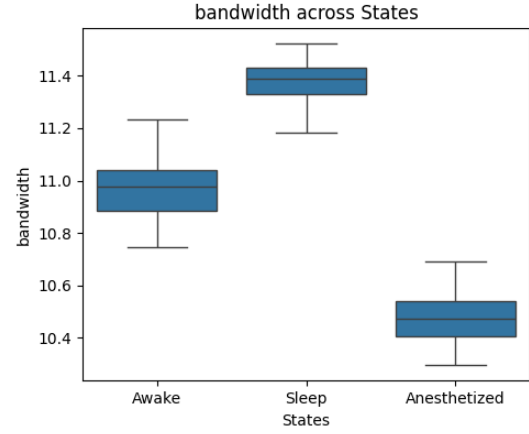
Experiment	State	ETC (mean $\pm$ SD)	Bandwidth (mean $\pm$ SD)	Max Scale (mean $\pm$ SD)	Fluctuation (mean $\pm$ SD)	Mean Entropy (mean $\pm$ SD)
KTMD Day 1	Anesthetized	23.91 $\pm$ 2.61	10.54 $\pm$ 1.74	33.05 $\pm$ 12.48	6.12 $\pm$ 2.35	0.01 $\pm$ 0.03
	Awake	27.27 $\pm$ 2.89	11.00 $\pm$ 1.50	18.26 $\pm$ 6.91	3.56 $\pm$ 1.42	0.12 $\pm$ 0.10
KTMD Day 2	Anesthetized	23.96 $\pm$ 2.74	10.41 $\pm$ 1.73	33.62 $\pm$ 12.67	6.21 $\pm$ 2.40	0.02 $\pm$ 0.03
	Awake	27.44 $\pm$ 2.98	10.93 $\pm$ 1.50	18.30 $\pm$ 6.68	3.59 $\pm$ 1.42	0.12 $\pm$ 0.10
SLP Day 1	Awake	26.49 $\pm$ 2.68	11.25 $\pm$ 1.45	19.77 $\pm$ 7.72	3.92 $\pm$ 1.60	0.10 $\pm$ 0.09
	Sleep	26.12 $\pm$ 2.74	11.36 $\pm$ 1.49	20.95 $\pm$ 8.65	4.10 $\pm$ 1.70	0.08 $\pm$ 0.08
SLP Day 2	Awake	26.34 $\pm$ 2.80	11.27 $\pm$ 1.47	20.43 $\pm$ 7.70	4.04 $\pm$ 1.58	0.09 $\pm$ 0.09
	Sleep	25.81 $\pm$ 2.68	11.40 $\pm$ 1.49	21.34 $\pm$ 8.18	4.15 $\pm$ 1.63	0.07 $\pm$ 0.08

6 Box Plot Visualisations

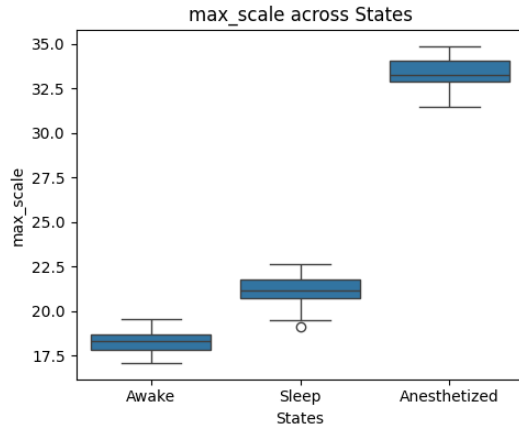
Box plots of each feature across the three consciousness states (Awake, Sleep, Anesthetized) are shown below.



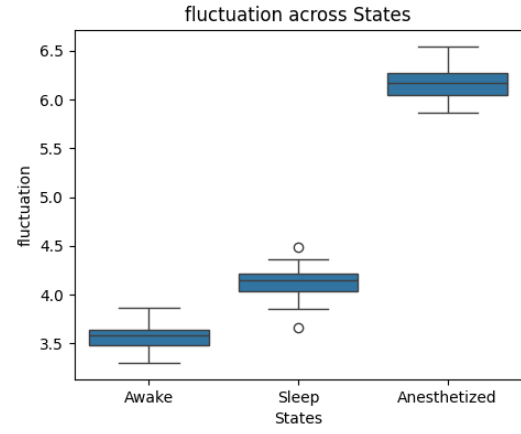
(a) ETC across states



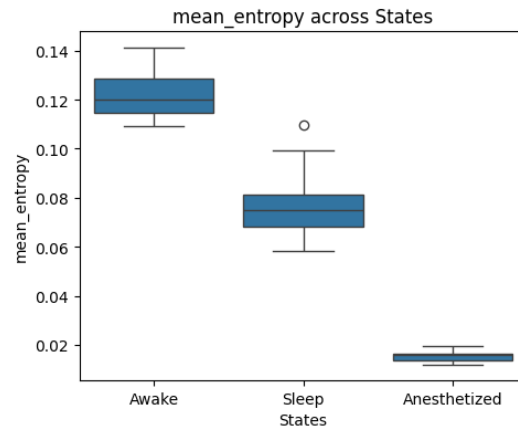
(b) Bandwidth across states



(c) Max Scale across states



(d) Fluctuation across states



(e) Mean Entropy across states

Figure 1: Box plots of all five features across Awake, Sleep, and Anesthetized states. Each box shows the interquartile range (IQR), with the median line inside. Whiskers extend to $1.5 \times \text{IQR}$; circles denote outliers.

7 Statistical Testing

7.1 Normality Test: Shapiro–Wilk

The Shapiro–Wilk test was applied to assess the normality of each feature across states. The test returns a state and a p -value.

Null Hypothesis H_0 : The data is normally distributed.
For $p > 0.05$, H_0 is accepted.

Table 6: Shapiro–Wilk Normality Test Results

Feature	Awake (p -value)	Sleep (p -value)	Anesthetized (p -value)	Normality Satisfied?
ETC	0.3354	0.0053	0.2654	No
Bandwidth	0.8605	0.8573	0.9289	Yes
Max Scale	0.9626	0.4324	0.5193	Yes
Fluctuation	0.9204	0.2726	0.8496	Yes
Mean Entropy	0.0979	0.0206	0.2350	No

Now, based on the normality results, the following tests are recommended:

Table 7: Recommended Statistical Tests per Feature

Feature	Recommended Test
Bandwidth	One-Way ANOVA
Max Scale	One-Way ANOVA
Fluctuation	One-Way ANOVA
ETC	Kruskal–Wallis
Mean Entropy	Kruskal–Wallis

8 Outlier Detection: Z-Score Test

A z-score outlier test identified the following:

Table 8: Outlier Identified via Z-Score Test

Index	Recording	ETC	Bandwidth	Max Scale	Fluctuation	Mean Entropy	ETC_z
1	Recording_02	26.93	11.18	19.08	3.66	0.11	2.95
3.00							

The corresponding p -value is:

$$p = 0.00533 < 0.05$$

which corresponds to recording02 under the sleep experiment

9 One-Way ANOVA Results

Null Hypothesis H_0 : The mean value of the particular feature is the same across all states.

9.1 Bandwidth

$$F\text{-statistic} = 635.32, \quad p\text{-value} = 1.24 \times 10^{-52}$$

9.2 Max Scale

$$F\text{-statistic} = 3456.20, \quad p\text{-value} = 1.29 \times 10^{-83}$$

9.3 Fluctuation

$$F\text{-statistic} = 2485.30, \quad p\text{-value} = 1.77 \times 10^{-77}$$

These results indicate that:

- The mean bandwidth differs significantly across Awake, Sleep, and Anesthetized states.
- Max Scale exhibits very large differences between consciousness states.
- Fluctuation differs significantly across the three states.

10 Kruskal–Wallis Test for ETC

Null Hypothesis H_0 : The ETC distributions are the same for Awake, Sleep, and Anesthetized states.

$$H\text{-statistic} = 79.03, \quad p\text{-value} = 6.89 \times 10^{-18}$$

Since $p \ll 0.05$, H_0 is rejected. ETC differs significantly across Awake, Sleep, and Anesthetized conditions.

10.1 Summary of Omnibus Tests

Table 9: Summary of Statistical Test Results

Feature	Test	Statistic	p -value
Bandwidth	ANOVA	$F = 635$	10^{-52}
Max Scale	ANOVA	$F = 3456$	10^{-83}
Fluctuation	ANOVA	$F = 2485$	10^{-77}
ETC	Kruskal–Wallis	$H = 79$	10^{-18}

11 Post Hoc Analysis

A post hoc analysis is used to identify which specific group pairs differ after a significant omnibus test (ANOVA or Kruskal–Wallis).

Table 10: Post Hoc Test Selection

Feature	Overall Test	Significant?	Post-Hoc Test
Bandwidth	ANOVA	Yes	Tukey HSD
Max Scale	ANOVA	Yes	Tukey HSD
Fluctuation	ANOVA	Yes	Tukey HSD
ETC	Kruskal–Wallis	Yes	Mann–Whitney + Holm

11.1 Tukey HSD Results

The Family-Wise Error Rate (FWER) is defined as:

$$\text{FWER} = P(\text{At least one Type I error among all tests})$$

FWER increases with the number of comparisons unless properly controlled.

11.1.1 Bandwidth

Table 11: Tukey HSD — Bandwidth Pairwise Comparisons

Comparison	Mean Diff.	Adj. p -value	95% CI Lower	95% CI Upper	Significant?
Anesthetized vs Awake	0.4905	< 0.001	0.4300	0.5510	Yes
Anesthetized vs Sleep	0.9031	< 0.001	0.8426	0.9636	Yes
Awake vs Sleep	0.4126	< 0.001	0.3521	0.4731	Yes

11.1.2 Max Scale

Table 12: Tukey HSD — Max Scale Pairwise Comparisons

Comparison	Mean Diff.	Adj. p -value	95% CI Lower	95% CI Upper	Significant?
Anesthetized vs Awake	−15.0626	< 0.001	−15.5213	−14.6038	Yes
Anesthetized vs Sleep	−12.1903	< 0.001	−12.6491	−11.7316	Yes
Awake vs Sleep	2.8722	< 0.001	2.4135	3.3309	Yes

Table 13: Tukey HSD — Fluctuation Pairwise Comparisons

Comparison	Mean Diff.	Adj. p -value	95% CI Lower	95% CI Upper	Significant?
Anesthetized vs Awake	−2.5880	< 0.001	−2.6803	−2.4957	Yes
Anesthetized vs Sleep	−2.0422	< 0.001	−2.1344	−1.9499	Yes
Awake vs Sleep	0.5459	< 0.001	0.4536	0.6381	Yes

Table 14: Column Abbreviations for Tukey HSD Tables

Column	Meaning
group1	First state
group2	Second state
meandiff	Difference in means
p-adj	Tukey-adjusted p -value
lower	Lower confidence interval bound
upper	Upper confidence interval bound
reject	Whether H_0 is rejected

11.1.3 Fluctuation

11.1.4 Column Abbreviations

11.1.5 State Ordering from Tukey HSD

Based on pairwise mean differences, the ordering of states for each feature is:

- **Bandwidth:** Sleep > Awake > Anesthetized
- **Max Scale:** Awake > Sleep > Anesthetized
- **Fluctuation:** Awake > Sleep > Anesthetized

One-way ANOVA revealed significant effects of consciousness state on bandwidth ($F = 635.32$, $p < 0.001$), maximum scale ($F = 3456.20$, $p < 0.001$), and fluctuation index ($F = 2485.30$, $p < 0.001$). Tukey’s HSD post-hoc analysis demonstrated significant pairwise differences among Awake, Sleep, and Anesthetized states for all three features (adjusted $p < 0.001$).

12 Mann–Whitney U Post Hoc Test for ETC

The Mann–Whitney U statistic is computed as:

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 \quad (1)$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2 \quad (2)$$

where n_1, n_2 are the sample sizes of the two groups, and R_1, R_2 are their respective rank sums. The final U -statistic is $\min(U_1, U_2)$.

12.1 Pairwise p -values (uncorrected)

$$\begin{aligned} \text{Awake vs Sleep: } p &= 3.34 \times 10^{-11} \\ \text{Awake vs Anesthetized: } p &= 3.02 \times 10^{-11} \\ \text{Sleep vs Anesthetized: } p &= 3.02 \times 10^{-11} \end{aligned}$$

12.2 Holm-Corrected p -values

After applying Holm correction:

$$p_{\text{corrected}} = [9.05 \times 10^{-11}, 9.05 \times 10^{-11}, 9.05 \times 10^{-11}]$$

13 Conclusion

Compression-spectrum-derived features — including ETC, bandwidth, maximum scale, and fluctuation index — exhibited statistically significant differences across Awake, Sleep, and Anesthetized states ($p < 0.001$), indicating their potential utility as markers of consciousness state.